DEPARTMENT OF MATHEMATICS
UNIVERSITY COLLEGE OF ENGINEERING (A) – O.U.,HYD - 7
B.E. (**AI&ML**) IV- SEMESTER, CLASS TEST- II

Engineering Mathematics – IV

(Numerical Methods)

1005237290008

TIME: 1hr 30 min **Max. Marks: 30**

	PART – A (Answer **all questions**) $5\times2=10$	Marks	CO	BT
1 a)	Explain Bisection method.	2	3	L1
b)	Find a real root of the equation, $x=e^{-x}$ using the Newton-Raphson method.	2	3	L2
c)	Find Lagrange interpolating polynomial for the following data x: 2, 2.5, 3.0 y: 0.69315, 0.91629, 1.09861	2	4	L2
d)	The distance (x cm) traversed by a particle at different times (t seconds) are given below. x: 0.0, 0.1, 0.2, 0.3 t: 3.01, 3.16, 3.29, 3.36 Find the velocity of the particle at t=0.3 seconds.	2	4	L3
e)	Write Runge-Kutta fourth order formula.	2	5	L1
	PART - B (Answer any **TWO** questions) $2\times10=20$			
2	**a.** Use Gauss elimination method to solve the system $2x+y+z=10,\ 3x+2y+3z=18,\ x+4y+9z=16$.	5	3	L3
	b. Solve the system $6x+y+z=20,\ x+4y-z=6,\ x-y+5z=7$ using LU decomposition method.	5	3	L3
3	**a.** From the table of cubes given below, find $(6.36)^3$ *and* $(6.61)^3$ x: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7 x^3: 226.981, 238.328, 250.047, 262.144, 274.625, 287.496, 300.763	4	4	L3
	b. Compute the values of $I=\int_0^1 \frac{dx}{1+x^2}$ using the trapezoidal rule with h=0.5 and using Simpson's 1/3 rule with h=0.25.	6	4	L3
4	**a.** Use Picard's method to obtain y(0.1) and y(0.2) of the problem defined by $\frac{dy}{dx}=x+yx^4,\ y(0)=3$.	5	5	L3
	b. Solve by Euler's method the problem $\frac{dy}{dx}=\frac{3}{5}x^3y,\ y(0)=1$.	5	5	L3

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